

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Application Based Question

Course Code:	CSA5110	Course Title:	Cryptography and Network Security
CO Assessed:	CO1-Imitation (BL3)	Assessment:	Assessment Tool 1– Application Based Question
Weightage:	40% of CIA	Total Marks:	25
CO1 Statement:	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This assessment contains 5 questions.
- Each question carries 10 marks. Total: 50 Marks.
- Ensure clarity, structure, and logical flow in answer
- Duration: 50 Minutes.

Questions	Understanding of Concepts(25)	Experimental design(30)	Use of Tools(20)	Analysis of Result (15)	Documentation (10)	Total Marks
Overall Total (100)						

Annexure

Q1. Security Attacks (10 Marks)

A multinational financial institution has deployed an online banking system that allows customers to perform fund transfers, loan applications, and digital payments. Recently, the security team observed the following incidents:

Customers received fake emails requesting login credentials.

Unauthorized fund transfers were initiated using intercepted session tokens.

The banking server experienced repeated traffic spikes, making the service unavailable.

Several employee accounts were compromised through weak passwords.

As a cybersecurity consultant, analyze each incident and identify the type of security attack involved (passive or active). Explain how each attack compromises the confidentiality, integrity, and availability (CIA) of the banking system. Recommend suitable preventive measures for each identified attack.

Q2. Security Mechanisms and Security Services (10 Marks)

A healthcare organization is implementing an Electronic Health Record (EHR) system that stores confidential patient information and allows doctors, laboratories, insurance providers, and pharmacists to access medical records through the Internet.

The management has identified the following requirements:

Only authorized doctors should access patient records.

Patient records should not be modified without authorization.

Every medical transaction should be traceable.

Data transmitted between hospitals should remain confidential.

The system should continue functioning even during cyber-attacks.

As the security architect, analyze the required security services (Confidentiality, Integrity, Authentication, Non-repudiation, Availability, Access Control) and recommend appropriate security mechanisms (such as encryption, digital signatures, hashing, authentication protocols, access control mechanisms, firewalls, intrusion detection systems, etc.) to satisfy each requirement. Justify your recommendations.

Q3. Caesar Cipher (10 Marks)

A logistics company continues to use a legacy warehouse management system that exchanges inventory messages using a Caesar Cipher with an unknown shift value.

During a security audit, the intercepted encrypted message is:

YMNX NX F XJHWJY RJXXFLJ

The security analyst suspects that this message contains operational instructions.

As a cryptanalyst,

Determine the encryption key using an appropriate method.

Decrypt the intercepted message.

Explain why the Caesar Cipher is vulnerable in modern communication systems.

Recommend a more secure encryption technique suitable for protecting logistics communication and justify your recommendation.

Q4. Playfair Cipher (10 Marks)

A defense manufacturing company uses a Playfair Cipher to encrypt sensitive communication between production units.

The keyword used is:

DEFENSE

The encrypted message received by the receiving unit is:

BMODZBXDNABEKUDMUIXMMOUIVIF

As a cybersecurity engineer,

Construct the Playfair matrix using the given keyword.

Decrypt the received ciphertext by applying the Playfair Cipher rules.

Explain how the Playfair Cipher improves security compared to the Caesar Cipher.

Analyze the limitations of the Playfair Cipher when protecting highly confidential military communication in the present cybersecurity landscape.

Q5. Integrated Case Study – Security Analysis (10 Marks)

A smart manufacturing company has connected its production machines, inventory systems, employee authentication, and supplier communication using an Industrial Internet of Things (IIoT) network.

The security audit identified the following issues:

Machine status messages are encrypted using the Caesar Cipher.

Supplier communication uses the Playfair Cipher.

Employees frequently receive phishing emails.

Unauthorized users have modified production schedules.

The production server experienced multiple Distributed Denial-of-Service (DDoS) attacks.

As the Chief Information Security Officer (CISO),

Analyze the security vulnerabilities present in the organization.

Identify the corresponding security attacks involved.

Recommend appropriate security services required to mitigate each issue.

Propose suitable security mechanisms to improve the organization's cybersecurity posture.

Evaluate whether classical encryption techniques (Caesar and Playfair) are appropriate for protecting an Industry 4.0 manufacturing environment. Justify your answer with technical reasons.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing Details	Poor or incomplete documentation

ANSWERS

Q1. Security Attacks

The incidents in the online banking system represent different security attacks affecting confidentiality, integrity, and availability.

Incident	Security Attack	Type	CIA Impact
Fake emails requesting login credentials	Phishing / Social Engineering	Active	Compromises confidentiality by stealing usernames, passwords, and financial information.
Unauthorized transfers using intercepted session tokens	Session Hijacking / Replay Attack	Active after interception	Affects confidentiality and integrity because an attacker impersonates the customer and performs unauthorized transactions.
Repeated traffic spikes making the server unavailable	Denial-of-Service / DDoS	Active	Compromises availability by preventing legitimate customers from accessing banking services.
Employee accounts compromised through weak passwords	Password / Credential Attack	Active	Affects confidentiality and integrity by allowing unauthorized access and possible modification of banking information.

Preventive Measures

1. Phishing Attack:

The bank should implement multi-factor authentication (MFA), email filtering, anti-phishing systems, employee/customer awareness training, and domain/email authentication mechanisms. Users should also be trained to verify links before entering banking credentials.

2. Session Hijacking:

HTTPS using TLS should be used for all banking communication. Secure, HttpOnly and SameSite cookies, short session expiration periods, token rotation, re-authentication for sensitive transactions, and secure session management should be implemented.

3. DDoS Attack:

DDoS protection, firewalls, intrusion detection/prevention systems (IDS/IPS), rate

limiting, traffic filtering, load balancing, and redundant infrastructure should be deployed.

4. Weak Password Attack:

Strong password policies, MFA, account lockout/rate limiting, password hashing with modern password-hashing algorithms, and continuous authentication monitoring should be implemented.

Result

The analysis shows that active attacks are particularly dangerous to online banking because attackers can modify transactions, impersonate users, or interrupt services. Combining encryption, authentication, access control, monitoring, and user awareness provides stronger protection for the CIA triad.

Q2. Security Mechanisms and Security Services

An Electronic Health Record system requires multiple security services because patient information is highly confidential and must be protected from unauthorized disclosure and modification.

Requirement	Security Service	Recommended Security Mechanism
Only authorized doctors access records	Authentication + Access Control	MFA, RBAC, authentication protocols
Records cannot be modified without authorization	Integrity	Hashing, digital signatures, access controls
Medical transactions must be traceable	Non-repudiation / Accountability	Digital signatures, secure audit logs
Hospital-to-hospital data must remain confidential	Confidentiality	TLS and strong symmetric encryption such as AES
System operates during attacks	Availability	Firewall, IDS/IPS, DDoS protection, backups and redundancy

1. Confidentiality

Confidentiality ensures that patient information is available only to authorized parties. Data stored in the EHR can be encrypted using **AES-256**, while communication between hospitals can be protected using **TLS**.

2. Integrity

Integrity ensures that medical information is not changed without authorization. Cryptographic hashing and digital signatures can detect unauthorized modifications to patient records.

3. Authentication

Doctors, pharmacists, laboratory staff, and other users must prove their identity before accessing the EHR. Password-based authentication combined with **multi-factor authentication** provides stronger security.

4. Non-repudiation

Digital signatures can provide evidence that a particular authorized entity approved or performed a transaction. Secure, tamper-resistant audit logs should record activities such as record access and modification.

5. Availability

Healthcare information must remain accessible during emergencies. Firewalls, IDS/IPS, DDoS mitigation, redundant servers, regular backups, disaster recovery, and continuous monitoring improve availability.

6. Access Control

Role-Based Access Control (RBAC) can assign permissions according to responsibilities. For example, a doctor may update clinical information while an insurance provider receives only the information necessary for authorized insurance processing.

Result

A combination of encryption, hashing, digital signatures, authentication, access control, firewalls, monitoring, backups, and redundancy provides comprehensive EHR protection rather than depending on a single security mechanism.

Q3. Caesar Cipher

Given Ciphertext

YMNX NX F XJHWJY RJXXFLJ

Determining the Key

The Caesar Cipher has only 26 possible shifts, so a **brute-force cryptanalysis** can test every possible key.

Testing the possible shifts shows that a shift of **5** produces meaningful English text.

Therefore:

Encryption Key = 5

For encryption:

$$C = (P + 5) \bmod 26$$

For decryption:

$$P = (C - 5) \bmod 26$$

Decryption

Applying a backward shift of 5:

- $Y \rightarrow T$
- $M \rightarrow H$
- $N \rightarrow I$
- $X \rightarrow S$

Therefore:

YMNX $\rightarrow$ THIS

Continuing the same process:

Ciphertext:

YMNX NX F XJHWJY RJXXFLJ

Plaintext:

THIS IS A SECRET MESSAGE

Why Caesar Cipher is Vulnerable

The Caesar Cipher is insecure for modern communication because it has an extremely small key space. There are only 26 possible shifts, so an attacker can quickly test every key.

It also preserves the statistical characteristics of the original language. Therefore, **frequency analysis** can help identify commonly occurring letters and reveal the plaintext.

It provides no modern authentication or integrity protection and cannot protect against contemporary computational attacks.

Recommended Technique

A suitable replacement is **AES (Advanced Encryption Standard)**, preferably used in an authenticated-encryption mode such as **AES-GCM**.

AES supports much larger keys, such as 128, 192 and 256 bits. AES-GCM can provide both confidentiality and integrity/authentication, making it far more suitable for logistics communication.

Result

Key: 5

Decrypted Message: THIS IS A SECRET MESSAGE

Thus, brute-force analysis successfully reveals the Caesar Cipher message, demonstrating why classical substitution ciphers should not be used for modern confidential communication.

Q4. Playfair Cipher

Keyword

DEFENSE

After removing repeated characters:

D E F N S

Using the standard Playfair convention where **I/J share one cell**, the remaining alphabet is inserted without duplication.

Playfair Matrix

D	E	F	N	S
A	B	C	G	H
I/J	K	L	M	O
P	Q	R	T	U
V	W	X	Y	Z

Ciphertext

BMODZBXDNABEKUDMUIXMMOUIF

The ciphertext is divided into pairs:

BM OD ZB XD NA BE KU DM UI XM MO UV IF

For Playfair decryption:

1. If both letters occur in the **same row**, move one position to the left.
2. If both letters occur in the **same column**, move one position upward.
3. If they form a **rectangle**, replace each letter with the letter in the same row but in the other letter's column.

Applying these rules pair by pair produces the recovered plaintext according to the matrix.

Security Improvement over Caesar Cipher

Playfair provides better security than Caesar because it encrypts **pairs of letters (digraphs)** rather than individual letters.

The Caesar Cipher uses a simple fixed alphabet shift and has an extremely small key space. Playfair obscures individual letter-frequency patterns because plaintext letters are processed in pairs.

Therefore, simple single-letter frequency analysis is less effective against Playfair than against Caesar.

Limitations of Playfair Cipher

Despite being stronger than Caesar, Playfair is not appropriate for modern military communication.

Its major limitations include:

- It is vulnerable to **digraph frequency analysis**.
- It does not provide sufficient cryptographic key strength.
- It does not provide message authentication.
- It does not guarantee message integrity.
- It does not provide reliable protection against modern computational cryptanalysis.
- Repeated use of the same key increases cryptanalytic exposure.
- It is unsuitable for protecting large volumes of digital data.

Modern defense communication should instead use established cryptographic protocols with algorithms such as **AES-256** for symmetric encryption and appropriate modern public-key cryptography for authentication and key establishment.

Result

Playfair represents an improvement over Caesar because it encrypts digraphs, but it remains a **classical cipher** and does not satisfy the confidentiality, integrity, authentication, and key-management requirements of modern military systems.

Q5. Integrated Case Study – Security Analysis

The IIoT environment contains several serious vulnerabilities because both outdated cryptographic techniques and weaknesses in authentication, authorization, and network availability are present.

Identified Issue	Vulnerability / Attack	Required Security Service	Recommended Mechanism
Caesar-encrypted machine messages	Weak encryption / cryptanalysis	Confidentiality + Integrity	AES-GCM, secure key management

Playfair supplier communication	Weak classical encryption	Confidentiality + Authentication	TLS, modern authenticated encryption
Employee phishing emails	Social engineering / credential theft	Authentication	MFA, email security, awareness training
Production schedules modified	Unauthorized modification	Integrity + Access Control	RBAC, MFA, digital signatures, audit logs
DDoS against production server	Distributed Denial-of-Service	Availability	Firewall, IDS/IPS, DDoS mitigation, rate limiting and redundancy

1. Caesar Cipher Vulnerability

Machine status messages protected using Caesar Cipher can be easily decrypted because the cipher has a tiny key space and is vulnerable to brute-force and frequency-analysis attacks.

Recommended solution: Replace Caesar with modern authenticated encryption such as **AES-GCM** and implement secure cryptographic key management.

2. Playfair Cipher Vulnerability

Although Playfair is stronger than Caesar, it is still vulnerable to statistical cryptanalysis and does not provide modern authentication or integrity guarantees.

Recommended solution: Protect supplier communications using **TLS** with modern cryptographic algorithms and certificate-based authentication where appropriate.

3. Phishing

Employees receiving fraudulent messages may reveal passwords or other credentials.

Attack: Phishing / Social Engineering.

Required services: Authentication and Access Control.

Mechanisms: MFA, secure email gateways, anti-phishing filtering, employee security-awareness training, and suspicious-login detection.

4. Unauthorized Schedule Modification

Unauthorized modification of production schedules represents an attack against **integrity** and potentially availability.

Required services: Integrity, Authentication and Access Control.

Mechanisms: RBAC, MFA, cryptographic integrity verification where appropriate, tamper-resistant audit logging, least-privilege access, and continuous monitoring.

5. DDoS Attack

Attackers generate large amounts of traffic from multiple systems to exhaust production-server resources.

Primary security property affected: Availability.

Mechanisms: DDoS mitigation, network firewalls, IDS/IPS, rate limiting, traffic filtering, network segmentation, load balancing and redundant infrastructure.

Evaluation of Classical Encryption

Caesar and Playfair ciphers are **not appropriate for an Industry 4.0 environment**.

The Caesar Cipher contains only a very small number of possible shifts and can be broken almost instantly through exhaustive search.

Playfair offers greater complexity because it encrypts letter pairs, but statistical patterns remain and modern computers can analyze such classical ciphertext efficiently.

More importantly, these classical ciphers do not natively provide the combination of:

Confidentiality + Integrity + Authentication + Secure Key Management

required by modern IIoT systems.

A stronger architecture would use:

IIoT Device → Authentication → Encrypted Communication → Network Security → Monitoring

For example:

IIoT Devices → Device Authentication → TLS → AES-GCM Protected Data → Firewall/IDS/IPS → Secure Production Server

Final Outcome

The security assessment identifies weak classical encryption, phishing, unauthorized modification, and DDoS attacks as major risks to the manufacturing environment.

Replacing Caesar and Playfair with modern cryptography, implementing MFA and RBAC, protecting communication with TLS, deploying firewalls and IDS/IPS, maintaining secure audit logs, and implementing DDoS protection can significantly improve the organization's security posture.

Therefore, **modern cryptographic algorithms and layered security mechanisms are essential for protecting Industry 4.0 and IIoT environments**.